

Analysis and Design

1. Module Specifications

1. User Authentication Module

- This module handles user registration, login, and logout functionality securely.

Inputs:

- Name
- Email
- Password

Processing:

- Password is encrypted using bcrypt
- User data stored in MongoDB
- Login credentials verified

Outputs:

- Successful login / error message

2. Dashboard Module

- Displays all users and allows selecting a user for chat.

Inputs:

- Logged-in user session

Processing:

- Fetch users from database
- Display list of users
- Handle user selection

Outputs:

- User list
- Selected chat window

3. Messaging Module

- Handles sending and receiving messages between users.

Inputs:

- Sender
- Receiver
- Message content

Processing:

- Message stored in database
- Retrieved for display

Outputs:

- Display of sent and received messages

4. Security Module (Hashing)

- Ensures message integrity using SHA-256 hashing.

Inputs:

- Message content

Processing:

- Generate hash using SHA-256
- Store hash with message
- Recalculate hash while displaying

Outputs:

- Message status:
- Secure ☒
- Tampered ☐

5. Settings Module

- Allows user to change UI theme and manage preferences.

Inputs:

- Theme selection (Light/Dark/Green)

Processing:

- Apply selected theme

Outputs:

- Updated UI appearance

6. Profile Module

- Allows user to view and update profile details.

Inputs:

- Name
- Email

Processing:

- Update user data in database

Outputs:

- Updated profile information

7. Real-Time Communication Module (Socket.IO)

- Handles real-time message updates.

Inputs:

- Message event

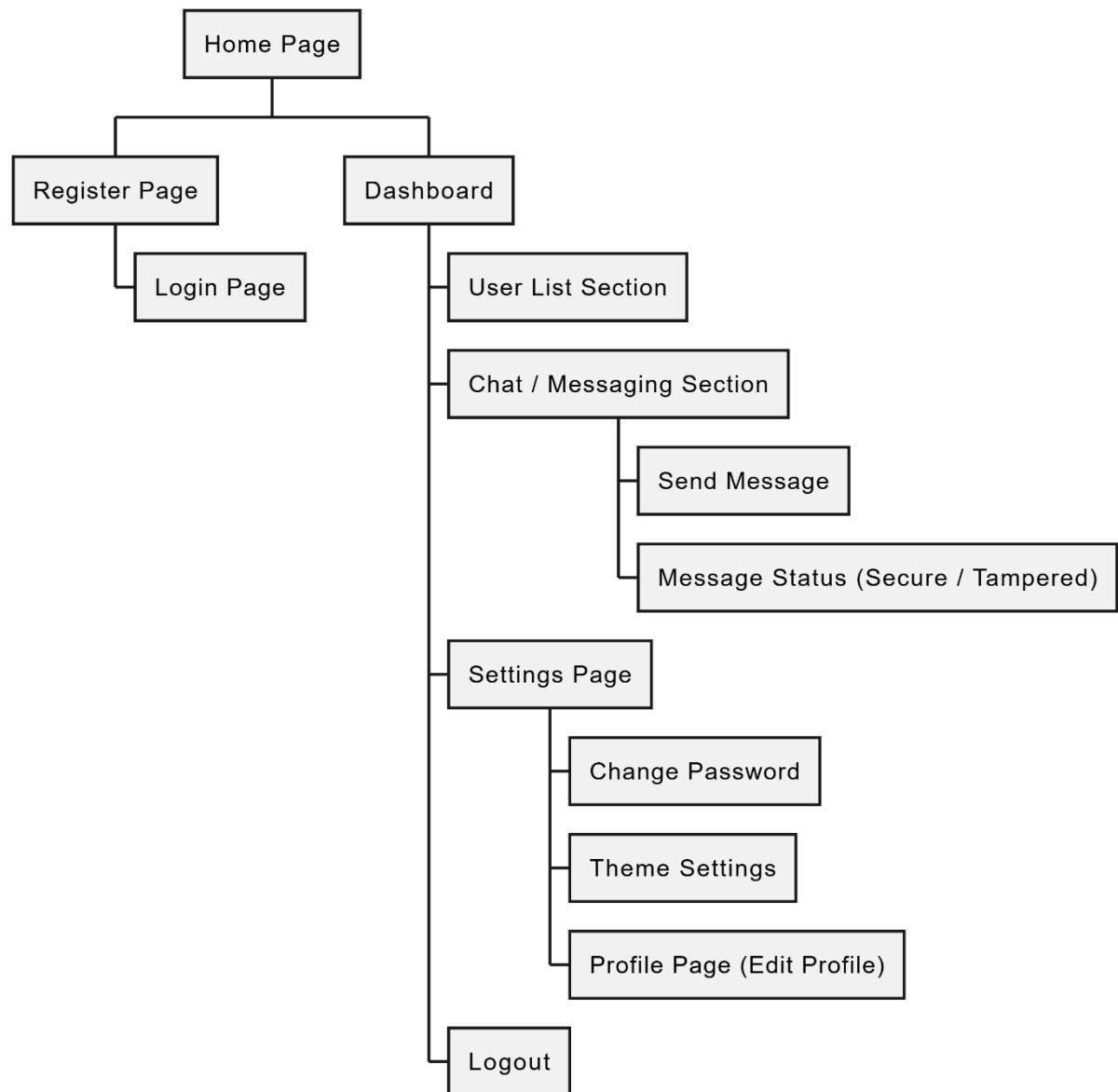
Processing:

- Emit and receive messages using sockets

Outputs:

- Instant message display

2. Website Map Diagram



Back End Specification

1. Normalization Database Design

Database normalization is the process of organizing data to reduce redundancy and improve efficiency. It helps keep data consistent and easy to manage by dividing it into related tables.

Unnormalized Form (UNF)

- Data is stored in an unstructured way.
- Multiple values in single field.
- Data redundancy present.

Unnormalized Form (UNF)

Sender Name	Sender Email	Receiver Name	Receiver Email	Message
Pragati	p@gmail.com	Rahul	r@gmail.com	Hi, Hello
Rahul	r@gmail.com	Pragati	p@gmail.com	Hey

First Normal Form (1NF)

- Atomic values ✓
- No repeating groups ✓

First Normal Form (1NF)

Sender Name	Sender Email	Receiver Name	Receiver Email	Message
Pragati	p@gmail.com	Rahul	r@gmail.com	Hi
Pragati	p@gmail.com	Rahul	r@gmail.com	Hello
Rahul	r@gmail.com	Pragati	p@gmail.com	Hey

Second Normal Form (2NF)

- Redundancy reduced
- Data separated
- Data split into multiple table

User table & Message table

Email	Name
p@gmail.com	Pragati
r@gmail.com	Rahul
c@gmail.com	Chaitali

Sender	Receiver	Message
p@gmail.com	r@gmail.com	Hi
p@gmail.com	r@gmail.com	Hello
r@gmail.com	p@gmail.com	HI

Sample Database Records:

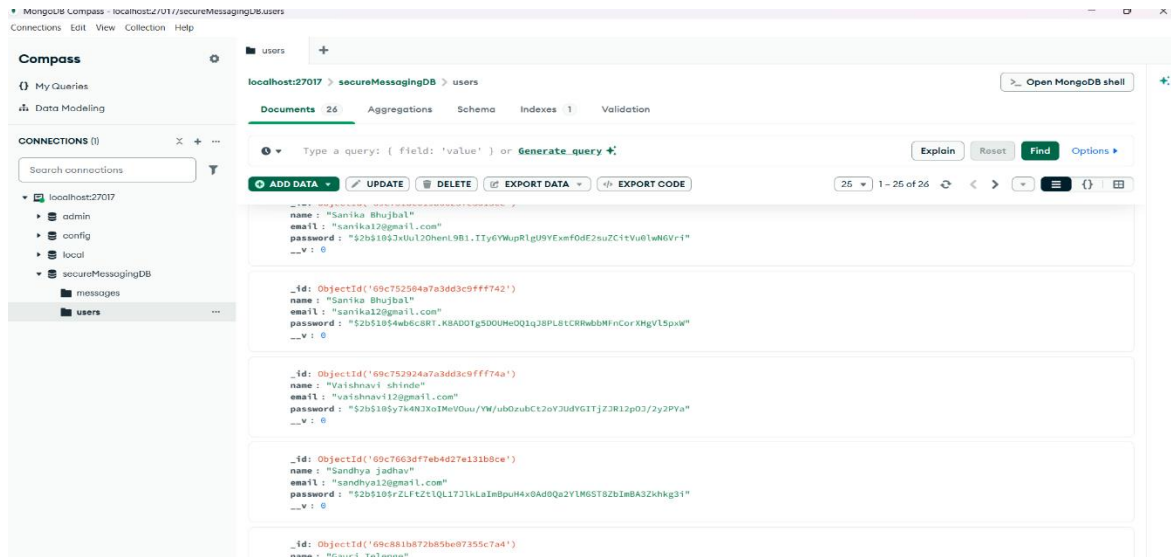
- Database: MongoDB (NoSQL Database)
- Stores data in JSON-like (BSON) document format
- Uses Collections and Documents instead of tables and rows
- Schema-less structure allows flexible data storage

User Collection

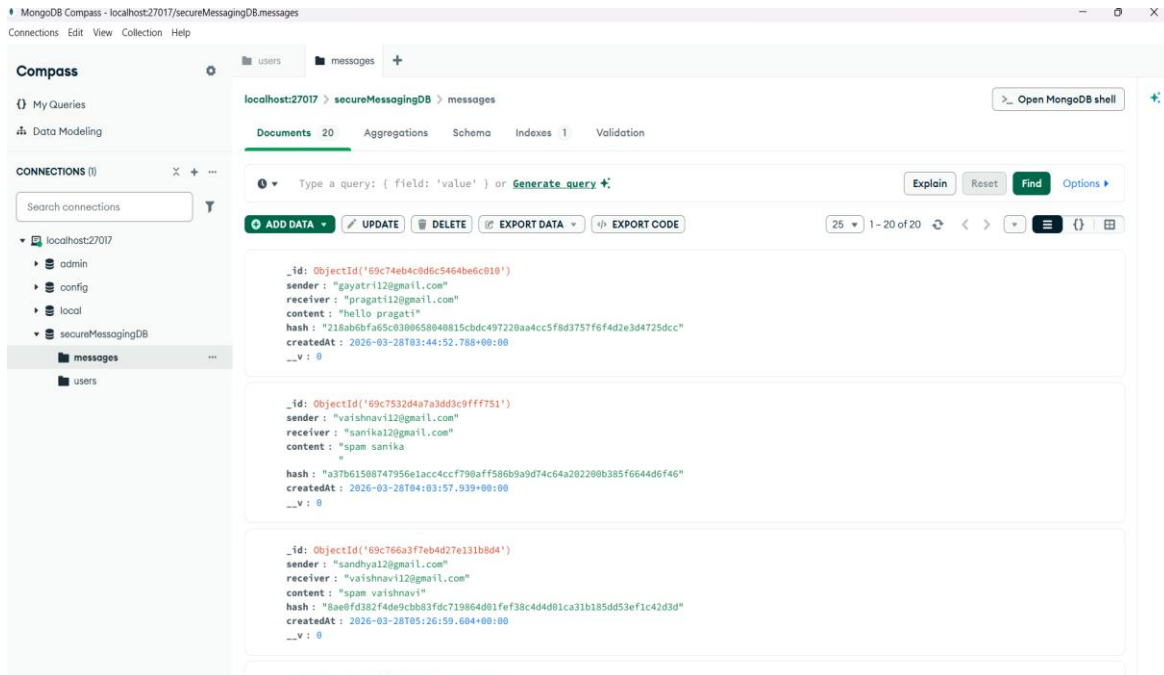
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the local host 'localhost:27017' and the selected database 'secureMessagingDB'. The 'users' collection is highlighted. The main panel shows the 'Documents' tab with 26 documents. A query filter is set to 'Type a query: { field: 'value' } or Generate query'. Below the filter, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. The document list shows four users:

- User 1:** _id: ObjectId('699ea6191e758ac731fdda85'), name: "pragati dasgude", email: "pragati12@gmail.com", password: "\$2b5185qF5tYEmxuY4m4HP4CMeYkuZK8aNIxDtbgJc5gCSIZNFenyLYyGCG"
- User 2:** _id: ObjectId('699eabd8ee2d89cbf78c5f7d'), name: "Gayatri", email: "gayatri12@gmail.com", password: "\$2b5185NgnvXpqnQF/VFoCTPPk1.eYK81a2AG6mDrv5Pwa1f72PsLV8h8Jt2"
- User 3:** _id: ObjectId('69c727368e069de2da9a6258'), name: "neha ghegade", email: "neha12@gmail.com", password: "\$2b51850DaY9CRIfrikwFmKHo3oJudInWa8S8BW3sXv81PteCx06kaVL0.."
- User 4:** _id: ObjectId('69c74f53c9d5c5464be6c814'), name: "Chaitali pacharne", email: "chaitali12@gmail.com", password: "\$2b51851btOGyJVV89..w9QhT0q220u85kpTgPm3v5EFH1b5ZAM..bCh3RBLXu"

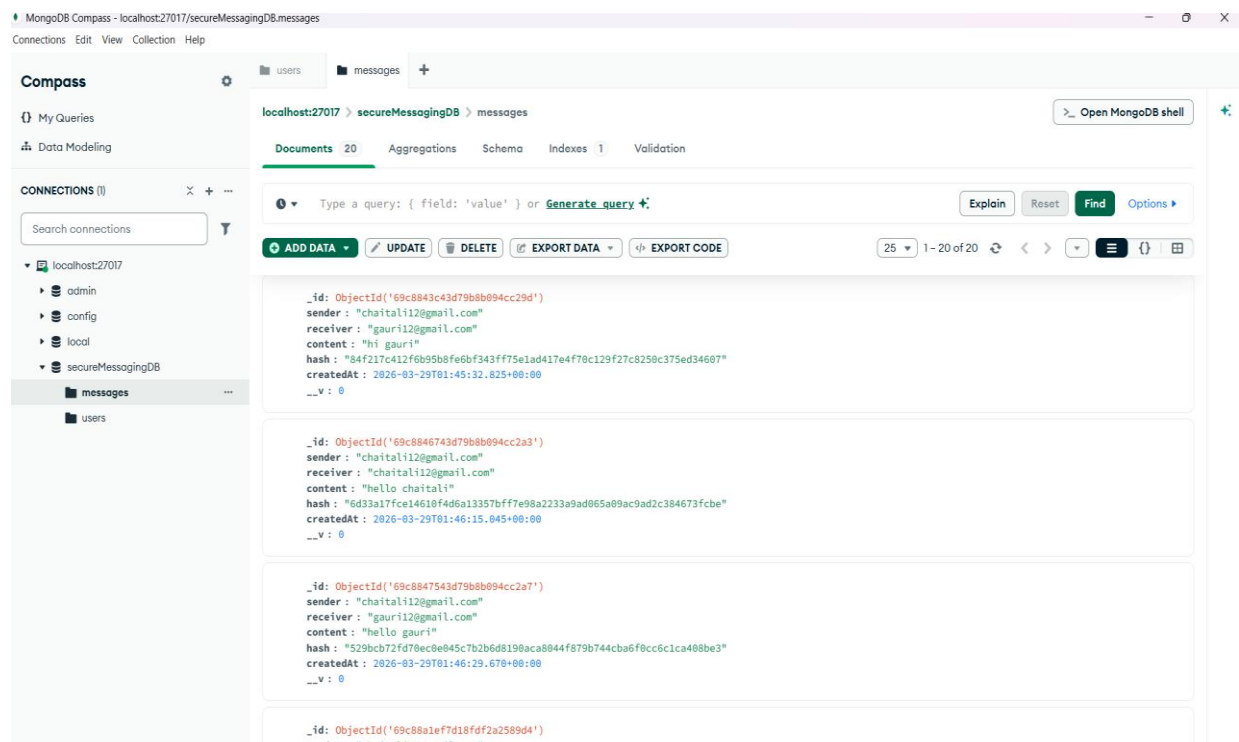
User Collection



Message Collection



Message Collection



2. Data Dictionary

A data dictionary is a collection of information about the data in a database. It describes each field, including its name, type, and purpose.

User Table

Field Name	Data Type
name	String
email	String
password	String

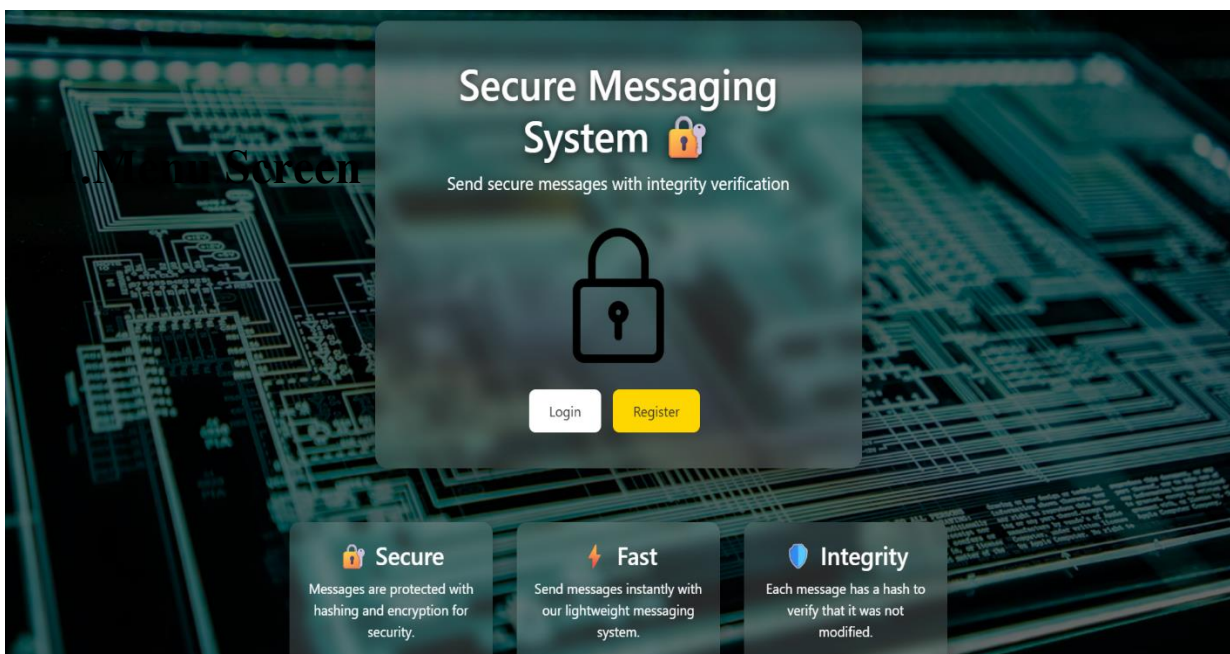
Message Table

Field Name	Data Type
sender	String
receiver	String
content	String
hash	String
createdAt	Date

Front End Specifications

1.Menu Screens

2.Input Screens



 Secure Register

Name

Email

Password

Register

Already have an account?

 Secure Login

Email

Password

Login

Don't have an account?

Users

Search user...

pragati12@gmail.com

gayatri12@gmail.com

neha12@gmail.com

chaitali12@gmail.com

sanika12@gmail.com

sanika12@gmail.com

vaishnavi12@gmail.com

sandhya12@gmail.com

gauri12@gmail.com

sakshi12@gmail.com

vaishnavi12@gmail.com

sonakshi12@gmail.com

snehal12@gmail.com

Secure Messaging Dashboard

Logged in as: pragati12@gmail.com

Logout

Send Secure Message

gayatri12@gmail.com

Type your message...

Send

Chat with: gayatri12@gmail.com

You

Message: hii gayatri

Status: ☒ Secure

29/3/2025, 12:28:25 pm

gayatri12@gmail.com

Settings

Change Password

New Password

Update Password

Theme Selector

Light **Dark** Green

Edit Profile Back to Dashboard

Secure Messaging Dashboard

Logged in as: pragati12@gmail.com

Logout

Send Secure Message

gayatri12@gmail.com

Type your message...

Send

Users

 Search user...

 pragati12@gmail.com

 gayatri12@gmail.com

 neha12@gmail.com

 chaitali12@gmail.com

 sanika12@gmail.com

 sanika12@gmail.com

 vaishnavi12@gmail.com

 sandhya12@gmail.com

 gauri12@gmail.com

 sakshi12@gmail.com

 vaishnavi12@gmail.com

 sonakshi12@gmail.com

 snehal12@gmail.com



Secure Messaging Dashb

Logged in as: **pragati12@gmail.com**

Logout

 **Edit Profile**

Name:

pragati dasgude

Email:

pragati12@gmail.com

Update Profile

Back to Dashboard

 Chat with: gayatri12@gmail.com

You

Message: hii gayatri

29/3/2026, 12:28:25 pm

Status:  Secure

gayatri12@gmail.com

Message: hello pragati

28/3/2026, 9:14:52 am

Status:  Secure

 **Secure Messaging Dashboard** 

Logged in as: neha12@gmail.com

Logout

 **Send Secure Message**

gauri12@gmail.com

Type your message...

Send

 Chat with: gauri12@gmail.com

gauri12@gmail.com

Message: spam neha

29/3/2026, 7:07:12 am

Status:  Tampered

Test Cases:

1. User Authentication Module

Test Plan:

- Verify user registration
- Validate login functionality
- Check error handling

Test Cases:

TC ID	Scenario	Input	Expected Output
TC01	Register valid user	Name, Email, Password	Registration successful
TC02	Register empty fields	Blank input	Error message
TC03	Login details	Correct email/password	Redirect to dashboard
TC04	Login wrong password	Incorrect password	Error message

2. Messaging Module

Test Plan:

- Check message sending
- Validate message storage
- Handle empty messages

Test Cases:

TC ID	Scenario	Input	Expected Output
TC05	Send message	Valid text	Message sent
TC06	Empty message	Blank	Not allowed
TC07	Receive message	Open chat	Message displayed

3. Dashboard Module

Test Plan:

- Verify user list display
- Check chat selection

Test Cases:

TC ID	Scenario	Input	Expected Output
TC08	View users	Open dashboard	User list shown
TC09	Select user	Click user	Chat opens

4. Settings & Profile Module

Test Plan:

- Verify password change
- Check profile update

Test Cases:

TC ID	Scenario	Input	Expected Output
TC10	Change password	Correct old + new	Updated successfully
TC11	Wrong old password	Incorrect input	Error message
TC12	Edit profile	New details	Profile updated

5. Security (Hashing) Module

Test Plan:

- Verify hash generation
- Check tampering detection

Test Cases:

TC ID	Scenario	Input	Expected Output
TC13	Original message	Normal message	Status: Secure
TC14	Tampered message	Modified data	Status: Tampered

Validation controls:

- Validation controls are used to ensure that the input provided by users is correct, complete, and secure. It helps in preventing invalid data entry and enhances the overall security and reliability of the system.

Validation Rules

- Email must be in a valid format (e.g., example@gmail.com)
- Password must be at least 6 characters long
- Message field cannot be empty
- Old password must be verified before changing password
- Invalid login credentials should display error message

Validation Table

Field Name	Validation Rule	Error Message
Name	Cannot be empty	“Name is required”
Email	Must be valid format	“Invalid email format”
Password	Minimum 6 characters	“Password too short”
Message	Cannot be empty	“Message cannot be empty”
Login	Correct credentials	“Invalid email or password”
Change Password	Old password must match	“Old password incorrect”

Source Code

Backend code:

```
JS server.js x
C:\Users\praga> OneDrive\ Desktop > secure-messaging-blockchain > JS server.js > ...

1  const express = require("express");
2  const path = require("path");
3  const mongoose = require("mongoose");
4  const bcrypt = require("bcrypt");
5  const crypto = require("crypto");
6  const Message = require("../models/Message");
7  const session = require("express-session");
8  const app = express();
9  const http = require("http");
10 const { Server } = require("socket.io");
11
12 const server = http.createServer(app);
13 const io = new Server(server);
14 const PORT = 3000;
15 // MongoDB Connection
16 mongoose.connect("mongodb://127.0.0.1:27017/secureMessagingDB")
17 .then(() => console.log("MongoDB connected"))
18 .catch(err => console.log(err));
19 app.use(express.static(path.join(__dirname, "public")));
20
21 app.get("/login", (req, res) => {
22   res.sendFile(path.join(__dirname, "public", "login.html"));
23 });
24
25 app.use(express.urlencoded({ extended: true }));
26
27 app.use(session({
28   secret: "mysecretkey",
29   resave: false,
30   saveUninitialized: false
31 }));
32
33 app.get("/register", (req, res) => {
34   res.sendFile(path.join(__dirname, "public", "register.html"));
35 });
36
37 app.post("/register", async (req, res) => {
```

```
JS server.js x
C:\Users\praga> OneDrive\ Desktop > secure-messaging-blockchain > JS server.js > ...

37 app.post("/register", async (req, res) => {
38   try {
39     const hashedPassword = await bcrypt.hash(req.body.password, 10);
40
41     const newUser = new User({
42       name: req.body.name,
43       email: req.body.email,
44       password: hashedPassword
45     });
46
47     await newUser.save();
48
49     res.redirect("/login");
50   } catch (error) {
51     console.log(error);
52     res.send("Error Registering User ✖");
53   }
54 });const userSchema = new mongoose.Schema({
55   name: String,
56   email: String,
57   password: String
58 });
59
60 const User = mongoose.model("User", userSchema);
61
62 app.post("/login", async (req, res) => {
63   try {
64     const user = await User.findOne({ email: req.body.email });
65
66     if (!user) {
67       return res.send("User Not Found ✖");
68     }
69
70     const isMatch = await bcrypt.compare(req.body.password, user.password);
71
72     if (isMatch) {
```

```
JS server.js
C:\Users\praga> OneDrive > Desktop > secure-messaging-blockchain > JS server.js > app.post("/login") callback
62 app.post("/login", async (req, res) => {
63
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69
70     const isMatch = await bcrypt.compare(req.body.password, user.password);
71
72     if (isMatch) {
73
74         req.session.user = user.email;
75
76         res.redirect("/dashboard");
77     } else {
78         res.send("Invalid Password ✖");
79     }
80
81     } catch (error) {
82         console.log(error);
83         res.send("Login Error ✖");
84     }
85 });
86
87 app.get("/dashboard", async (req, res) => {
88     if (!req.session.user) {
89         return res.redirect("/login");
90     }
91     const selectedUser = req.query.chat;
92     let messages = [];
93
94     if (selectedUser) {
95         messages = await Message.find({
96             $or: [
97                 { sender: req.session.user, receiver: selectedUser },
98                 { sender: selectedUser, receiver: req.session.user }
99             ]
100         }).sort({ createdAt: -1 });
101     }
102     const users = await User.find();
103     let html = `
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Frontend Code

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index.html X
C:\Users\praga>OneDrive\Desktop>secure-messaging-blockchain>public>index.html>...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Secure Messaging</title>
5 <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">
6 <style>
7 body{
8   font-family: 'Poppins', sans-serif;
9   background: linear-gradient(rgba(0,0,0,0.6), rgba(0,0,0,0.6)),
10    url('https://images.unsplash.com/photo-1550751827-4bd374c3f58b?auto=format&fit=crop&w=1400&q=80');
11   background-size: cover;
12   background-position: center;
13   height: 100vh;
14 }
15 </style>
16 <style>
17 body{
18   font-family: 'Poppins', sans-serif;
19   background: linear-gradient(rgba(0,0,0,0.6), rgba(0,0,0,0.6)),
20    url('https://images.unsplash.com/photo-1550751827-4bd374c3f58b?auto=format&fit=crop&w=1400&q=80');
21   background-size: cover;
22   background-position: center;
23   height: 100vh;
24 }
25 /* Button style */
26 .btn-custom{
27   padding:12px 25px;
28   border-radius:8px;
29   text-decoration:none;
30   margin-right:10px;
31   display:inline-block;
32   transition:0.3s;
33 }
34
35 /* login button */
36 .login-btn{
37   background: white;
38 }
39
40 Downloading update: 63.4 MB / 133.1 MB + 48%
```

```
File Edit Selection View Go Run Terminal Help
index.html X login.html X
C:\Users\praga>OneDrive\Desktop>secure-messaging-blockchain>public>login.html>...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Secure Login</title>
5
6 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
7
8 <style>
9 body{
10   background:linear-gradient(135deg, #0f2027, #203a43, #2c5364);
11   height:100vh;
12   display:flex;
13   justify-content:center;
14   align-items:center;
15   font-family:Arial;
16 }
17
18 .login-card{
19   width:380px;
20   background:#111;
21   color:white;
22   padding:30px;
23   border-radius:15px;
24   box-shadow:0 0 20px rgba(0,255,255,0.3);
25 }
26
27 .login-card h3{
28   text-align:center;
29   margin-bottom:25px;
30   color:#00ffff;
31 }
32
33 .form-control{
34   background:#222;
35   border:none;
36   color:white;
37 }
38
39 Downloading update: 63.8 MB / 133.1 MB + 48%
```

```
File Edit Selection View Go Run Terminal Help
server.js index.html login.html settings.html X
C:\Users\praga>OneDrive>Desktop>secure-messaging-blockchain>public>settings.html>...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Settings</title>
5 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
6 <style>
7 body { transition: background 0.3s; }
8 .theme-light { background: #faf9fa; color: #000; }
9 .theme-dark { background: #121212; color: #fff; }
10 .theme-green { background: #e6f4ea; color: #000; }
11 </style>
12 </head>
13 <body class="theme-light">
14 <div class="container mt-5">
15 <h2>Settings</h2>
16 <hr>
17
18 <h3>Change Password</h3>
19 <form action="/change-password" method="POST">
20 <div class="mb-3">
21 <input type="password" class="form-control" name="oldPassword" placeholder="Old Password" required>
22 </div>
23 <div class="mb-3">
24 <input type="password" class="form-control" name="newPassword" placeholder="New Password" required>
25 </div>
26 <button type="submit" class="btn btn-primary">Update Password</button>
27 </form>
28
29 <hr>
30 <h3>Theme Selector</h3>
31 <div class="mb-3">
32 <button class="btn btn-light" onclick="changeTheme('light')">Light</button>
33 <button class="btn btn-dark" onclick="changeTheme('dark')">Dark</button>
34 <button class="btn btn-success" onclick="changeTheme('green')">Green</button>
35 </div>
36
37 <hr>
38
39 </div>
40 </body>
41 </html>
42
Download update: 64.1 MB / 133.1 MB • 48%
Ln 1, Col 1 Spaces: 2 UTF-8 CRLF [ ] HTML
```

```
File Edit Selection View Go Run Terminal Help
server.js index.html login.html settings.html register.html X
C:\Users\praga>OneDrive>Desktop>secure-messaging-blockchain>public>register.html>...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Secure Register</title>
5
6 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
7
8 <style>
9
10 body{
11 background:linear-gradient(135deg,#0f2027,#203a43,#2c5364);
12 height:100vh;
13 display:flex;
14 justify-content:center;
15 align-items:center;
16 font-family:Arial;
17 }
18
19 .register-card{
20 width:380px;
21 background:#111;
22 color:#fff;
23 padding:30px;
24 border-radius:15px;
25 box-shadow:0 0 20px rgba(0,255,255,0.3);
26 }
27
28 .register-card h3{
29 text-align:center;
30 margin-bottom:25px;
31 color:#00ffff;
32 }
33
34 .form-control{
35 background:#222;
36 border:none;
37 color:#fff;
38 }
39
40 </style>
41
42 </head>
43 <body>
44 <div class="register-card">
45 <h3>Secure Register</h3>
46 <div class="form-control">
47 <input type="text" placeholder="Name" required>
48 </div>
49 <div class="form-control">
50 <input type="password" placeholder="Password" required>
51 </div>
52 <div class="form-control">
53 <input type="password" placeholder="Confirm Password" required>
54 </div>
55 <button type="submit" class="btn btn-primary">Register</button>
56 </div>
57 </body>
58 </html>
59
Download update: 64.3 MB / 133.1 MB • 48%
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF [ ] HTML
```

User Manual

1. Open the application
2. Register a new user
3. Login with credentials
4. Select a user from dashboard
5. Send message
6. View message status (Secure/Tampered)
7. Change settings if needed
8. Logout